Internal Flow and Heat Transfer in a Centrifugal Pump

~23BME013

Aim: To study centrifugal pressure rise, absolute and relative velocities, pump head, and temperature change in a simplified rotating annular passage.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cylindrical
Grid points, along radius	80
Grid points along theta	160
Reynold's number	500
Total Iterations	1200

Governing Equation:

The Leading-order centrifugal pressure rise is:

$$\Delta p \approx \frac{1}{2}\rho\Omega^2(r_2^2 - r_1^2).$$

Thermal transport is represented by:

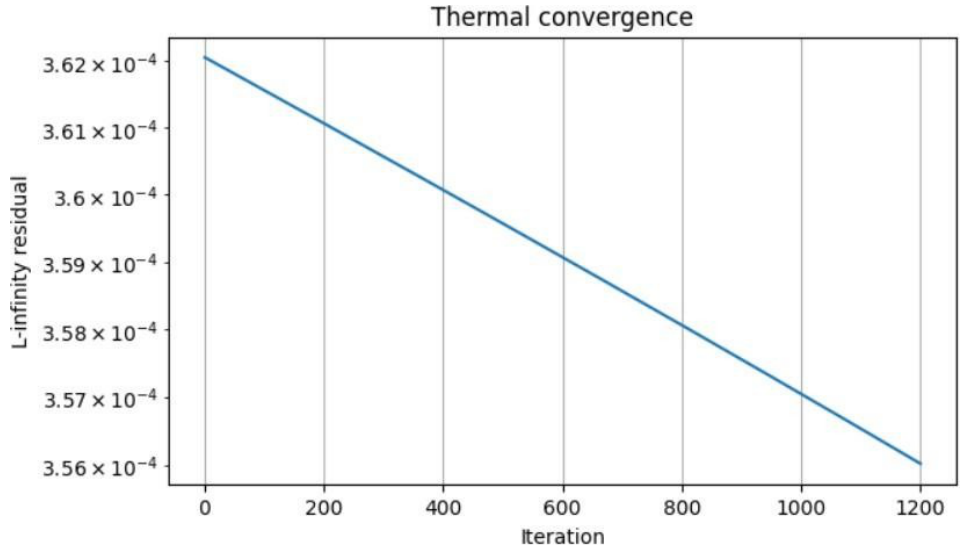
$$\rho c_p \, \mathbf{u}_{rel} \cdot \nabla T = k\nabla^2 T + S_T.$$

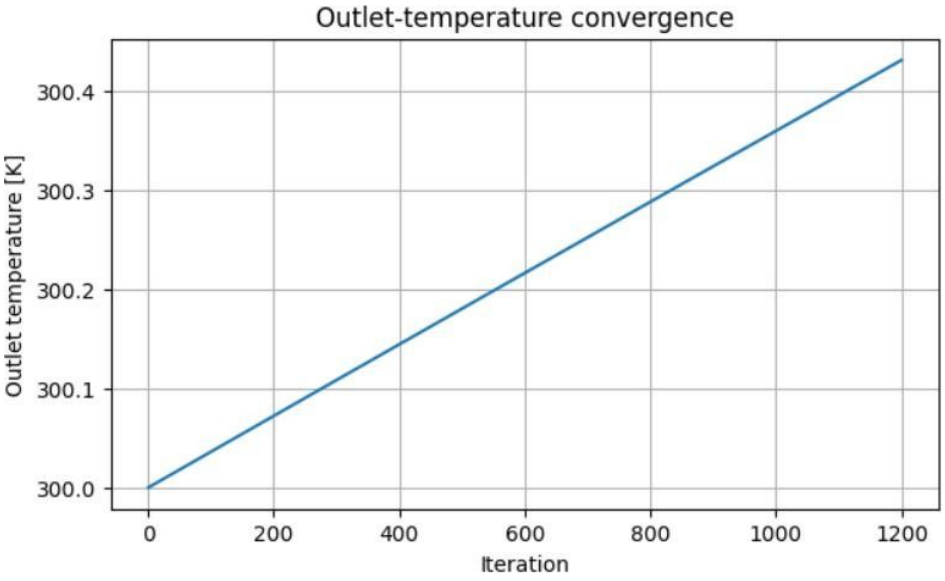
Boundary Conditions: Specified inlet temperature and radial inflow at the eye; outward radial transport; periodic azimuthal direction; wall heat exchange represented through a distributed source.

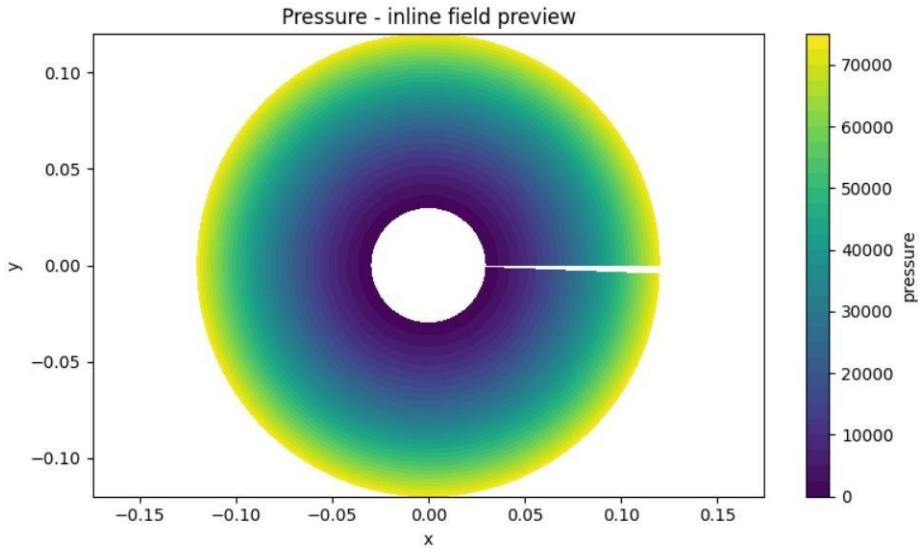
Numerical Method: Polar single-rotating-frame reduced-order model with prescribed radial and tangential velocity components, centrifugal pressure rise, wall heat transfer, and loss heating.

Observations: Pressure rise, pump head, inlet and outlet temperatures, maximum absolute speed, relative velocity, and thermal residual at various rpm of pump and constant Raynold's number.

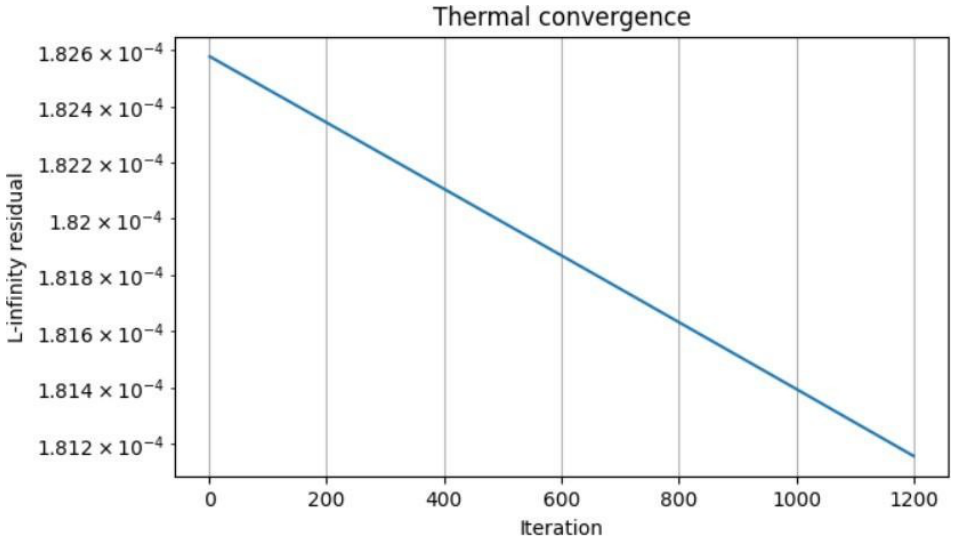
For pump rotation at 1013 rpm:

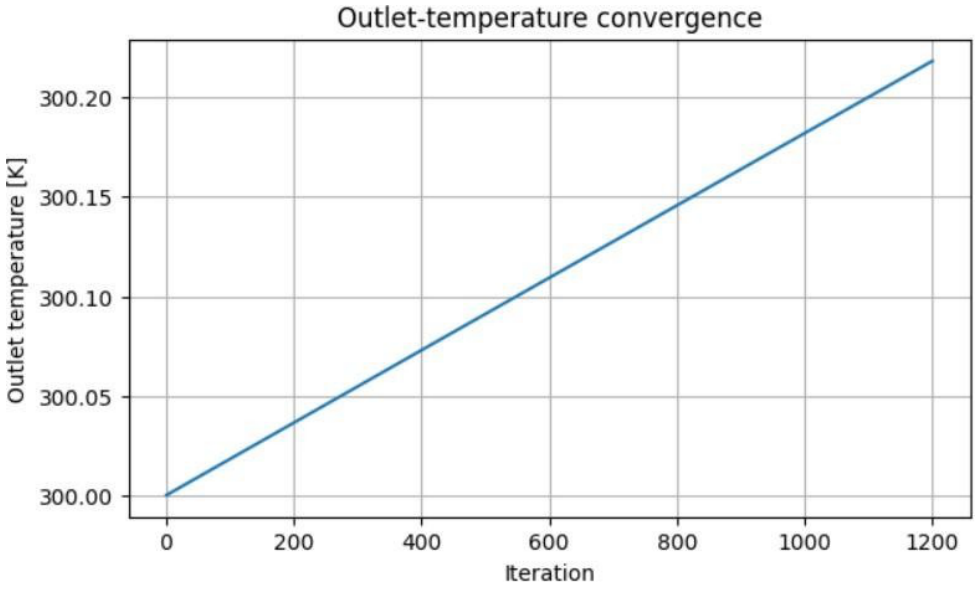


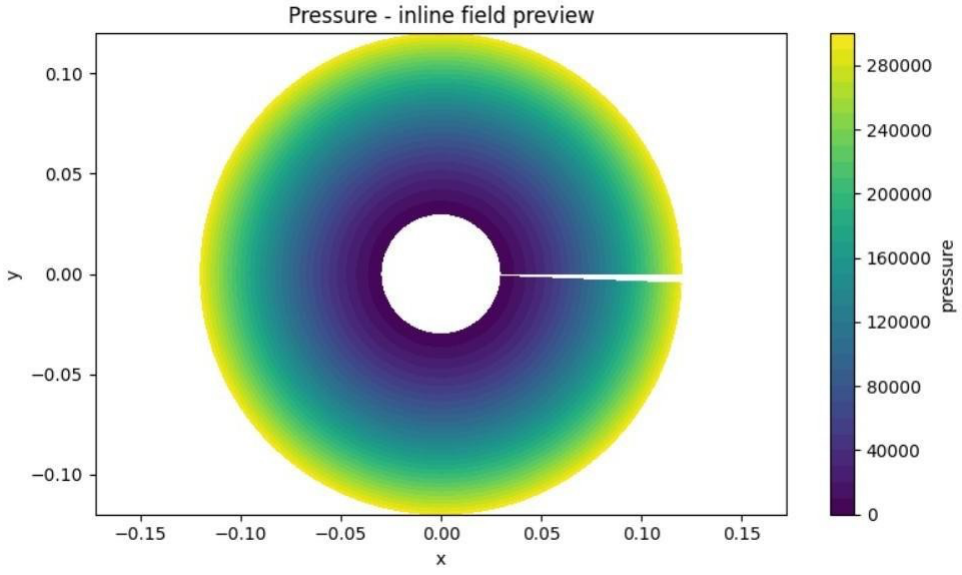




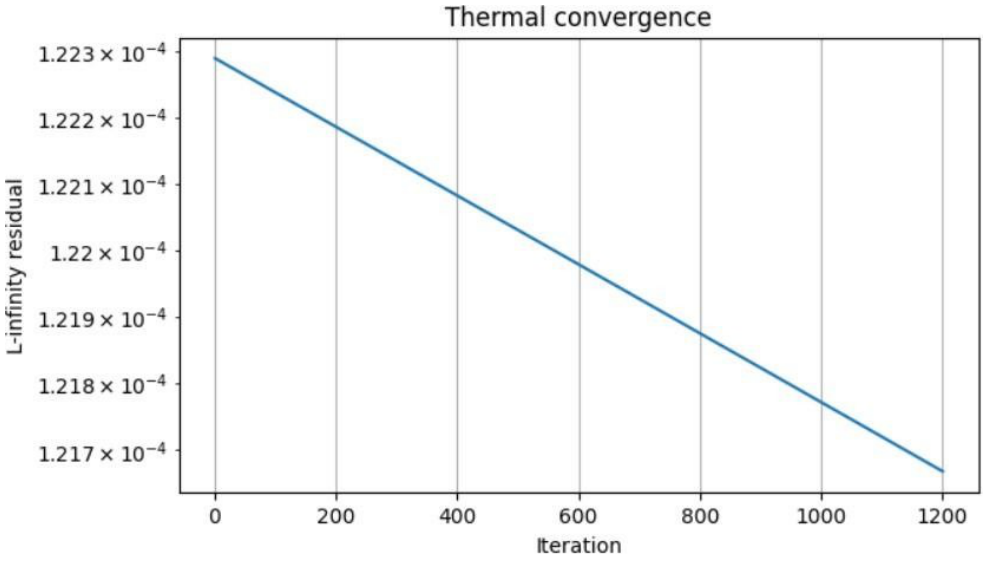
For pump rotation at 2013 rpm:

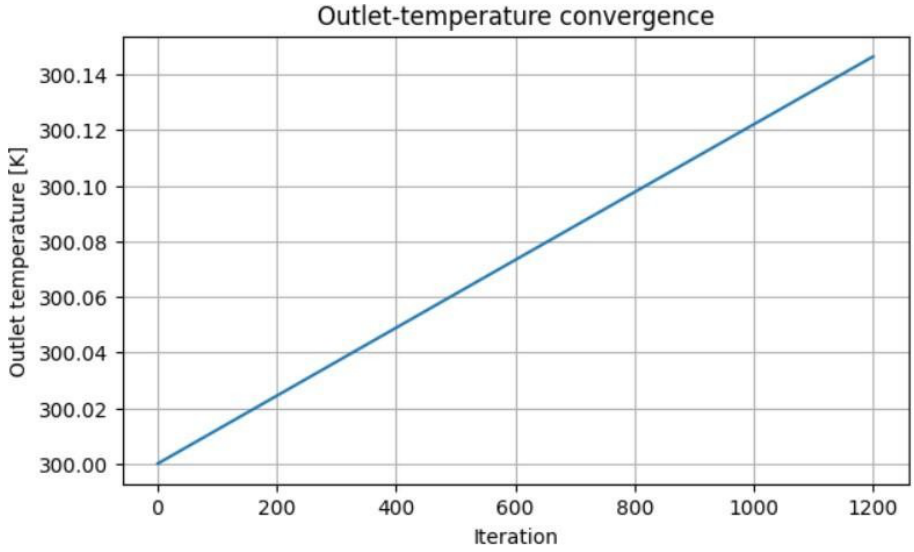


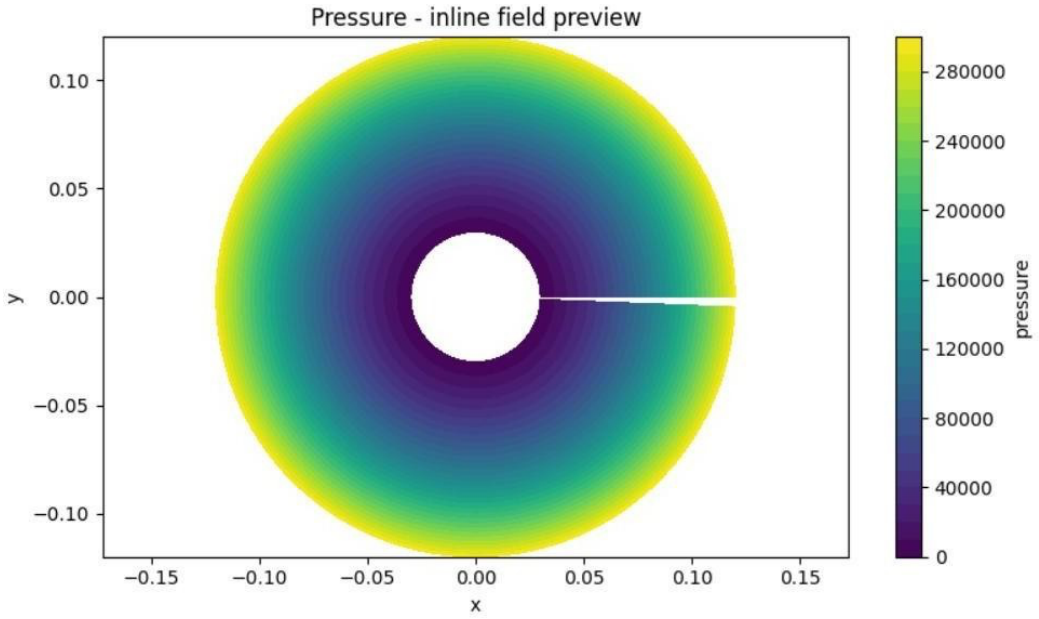

Outlet-temperature convergence
Outlet temperature [K]
300.20
300.15
300.10
300.05
300.00
0
200
400
600
800
1000
1200
Iteration


Pressure - inline field preview
0.10
0.05
0.00
−0.05
−0.10
y
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
x
280000
240000
200000
160000
120000
80000
40000
0
pressure

For pump rotation at 3013 rpm:







Result:

Pump Speed (RPM)	Reynold's number	Inlet temperature (k)	Outlet temperature (k)	Thermal residual	Max pressure from contour
1013	500	300	300.3	$3.56x10^{-4}$	$7x10^{4}$
2013	500	300	300.18	$1.81x10^{-4}$	$2.8x10^{5}$
3013	500	300	300.12	$1.22x10^{-4}$	$2.8x10^{5}$

Conclusion:

The centrifugal pump simulations were performed at 1013, 2013, and 3013 rpm while maintaining a constant Reynolds number of 500. The inlet temperature was maintained at approximately 300 K for all cases. At 1000 rpm, the outlet temperature increases to approximately 300.30 K, corresponding to a temperature rise of approximately 0.30 K. Increasing the rotational speed to 2000 rpm results in an outlet temperature of approximately 300.18 K, while at 3013 rpm, the outlet temperature approaches approximately 300.12 K. The corresponding thermal residuals decrease from approximately 3.56×10^{-4} at 1013 rpm to 1.81×10^{-4} at 2013 rpm and 1.22×10^{-4} at 3013 rpm. The pressure field shows a substantial increase in centrifugal pressure rise with rotational speed, with the pressure scale increasing from approximately 7×10^{4} at 1013 rpm to approximately 2.8×10^{5} at the higher rotational-speed cases. These results demonstrate the strong influence of rotational speed on centrifugal pressure generation, while the predicted temperature variation remains relatively small, with the maximum observed temperature increase being approximately 0.30 K.